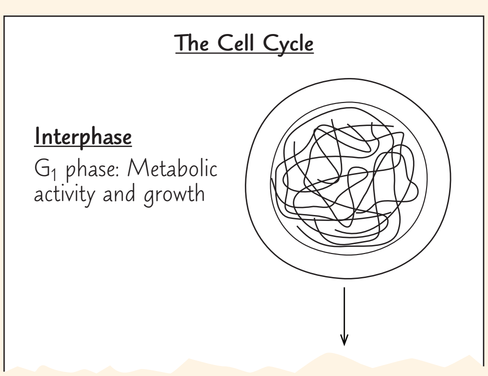
12 The Cell Cycle

KEY CONCEPTS

Study Tip

Make a visual study guide: Figure 12.1 presents the events of the cell cycle as a simplified linear diagram. As you learn more about the cell cycle, draw a detailed linear diagram of the stages of interphase, mitosis, and cytokinesis. Include explanatory labels. Add the circular chart from Figure 12.6 and show how the two diagrams are related.



Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 12
- BioFlix® Animation: Mitosis
- Animation: Microtubule Depolymerization

For Instructors to Assign (in Item Library)
- BioFlix® Tutorial: Mitosis (2 of 3): Mechanisms of Mitosis
- Coaching Activity: Evaluating Science in the Media: Tanning and Skin Cancer

Ready-to-Go Teaching Module (in Instructor Resources)
- Mitosis (Concept 12.2)

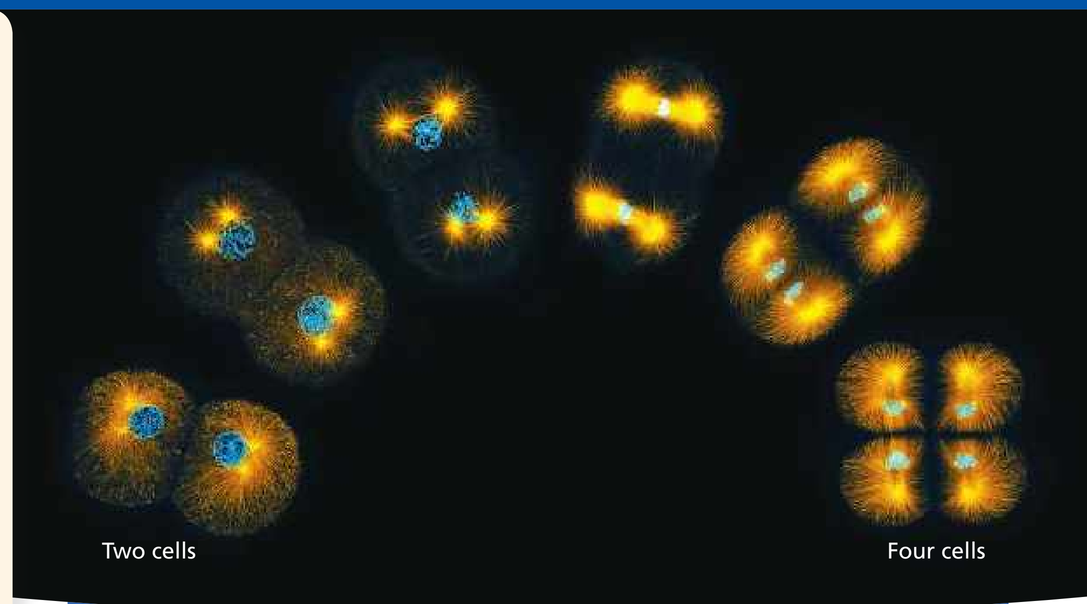


Figure 12.1 A multicellular organism starts out as a single cell that divides into two. Those two cells then divide into four, as shown in these fluorescence micrographs of a marine worm embryo. Cell division continues throughout an organism's life, for growth or to replace worn-out or damaged cells. Each time a cell divides in this way, it is crucial that the daughter cells be genetically identical to the parent cell.

How does one parent cell give rise to two genetically identical daughter cells?